

ESDNet: Epilepsy Seizure Detection using a Novel Hybrid Deep Sequential Network from EEG Signals

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Abstract—Epilepsy is a neurological disorder typically characterized by recurrent, sudden seizures. Seizures in patients with epilepsy often result in involuntary movements and fainting, which can lead to severe injuries or even death in certain situations. Manual seizure detection depends on long-duration electroencephalogram (EEG) signal inspection, which is computationally and labor intensive and necessitates expert involvement. Therefore, early and automated epilepsy detection from EEG signals is an important task for system biologists and neurologists to ensure the safety of epileptic patients throughout a seizure and prevent possible dangers as earliest. A novel Hybrid Deep Sequential Network for Epilepsy Seizure Detection from EEG Signals (ESDNet) is proposed in this article. It utilizes the benefits of One-dimensional Convolutional Neural Network (1D CNN) and Bidirectional Gated Recurrent Unit (BiGRU) networks as it combines the 1D CNN with BiGRU deep sequential architecture. Here, three layers of 1D CNN and two layers of BiGRU are used for extracting features from EEG sequences and a dense layer is used as a classifier. The performance of the proposed ESDNet model is validated on two EEG datasets viz., UCI Epileptic Seizure Dataset and CHB-MIT Scalp EEG Dataset with respect to precision, recall, F_1 -score measure, accuracy, confidence interval and its result is compared with existing state-of-the-art methods. In the UCI Epileptic Seizure Dataset and CHB-MIT Scalp EEG Dataset, the proposed ESDNet model achieved the accuracy rate of 99.74% and 93.20% respectively compare to existing methods.

Keywords—Epilepsy, Seizure, EEG Signal, Deep Sequential, 1D CNN, BiGRU.

I. INTRODUCTION

Epilepsy is a brain disorder that afflicts people of all ages. It occurs when there is an erratic activity in the brain's neurons. This can lead to seizures, which are surges of neuroelectricity in the brain. These seizures can vary in type and severity impacting daily life in different ways. Roughly 50 million people are affected by epilepsy across the globe, and 2.4 million people throughout the world suffer from epilepsy each year, as per the report of the WHO. Epileptic patients have a two to three fold higher risk of early death compared to the average healthy individual. People with epilepsy may experience seizures that involve staring blankly or jerking of the arms or legs for a few moments. An epilepsy diagnosis often requires a minimum of two seizure attacks without any

identifiable stimulus. Hence, it is an important area of research for neurologists and system biologists to automatically recognize epileptic seizure for early prediction and seizure disorder diagnosis.

Electroencephalography (EEG) is widely implemented to identify epileptic seizures. It is a non-invasive method that measures voltage changes in neurons due to ionic movements, providing a comprehensive representation of the electrophysiological brain activity. EEG recording is a valuable diagnostic tool for analyzing brain functions and detecting epilepsy seizures. Despite technological advancements, the process of continuous observation of EEGs to monitor epileptic patients remains time-consuming for neurologists and usually costly for epileptic patients. Computational biologists have developed methods to assist neurologists in detecting epilepsy in patients at the earliest.

Researchers have conducted several studies to identify seizures by using machine-learning methods that analyze EEG signals. The difficulty with machine learning (ML) approaches is that categorisation cannot be done directly from raw inputs. Prior to classification, preprocessing is a step in the feature extraction and feature selection procedures. In seizure-detection systems, these preprocessing techniques increase overhead. Raw signals can be inputted into deep learning (DL) models without feature extraction or selection [1].

The major contributions are outlined below:

- A novel Hybrid Deep Sequential Network for Epilepsy Seizure Detection from EEG Signals (ESDNet) is proposed in this study.
- The proposed ESDNet model is an amalgamation of One-dimensional Convolutional Neural Network (1D CNN) and Bidirectional Gated Recurrent Unit (BiGRU) deep sequential learning framework. The 1D CNN is a highly effective, noise-resistant, and computationally efficient method for identifying significant features from EEG signals. It is well-suited for different types of neurological and cognitive studies. Meanwhile, the BiGRU captures dependencies both past and future time steps, manages long-term patterns, optimizes memory usage, and enhances accuracy in sequential datasets.

- The proposed ESDNet model is extensively validated on two benchmark EEG datasets using standard classification performance indicators.

The structure of this article is outlined below. Section II discusses the related works, while the materials and methods are detailed in Section III. Experimental results and compared methods are presented in Sections IV and V, respectively. Section VI provides concluding insights and outlines future works.

II. RELATED WORKS

A review of relevant studies on EEG-based classification approaches is presented in this section.

A deep metric learning-based seizure detection model using one-dimensional convolutional EEG representations was reported by L. Duan *et al.* [2], achieving 86.68% accuracy on the CHB-MIT Scalp EEG Dataset. J. Zhang *et al.* [3] presents a novel method using Temporal Convolutional Neural Networks (TCNs) for EEG signal classification. V. Leelavathi and M. Sambath [4] used CNN and bidirectional LSTM algorithms for seizure prediction from EEG data. The Bi-LSTM model outperforms CNN, achieving over 98.15% accuracy in classifying seizure activity, highlighting its effectiveness in capturing temporal dependencies in EEG recordings. Ayesha *et al.* [5] developed a ML model to sort EEG signals into two categories: ictal (related to seizures) and interictal (between seizures) and used both fuzzy-based and traditional algorithms. F. Hassan *et al.* [6] introduced a hybrid approach that combines CNN for feature extraction and ML classifiers. O. Kaziha *et al.* [7] designed a CNN for automatic seizure recognition from raw EEG signals on the CHB-MIT dataset, eliminating the requirement for manual feature extraction and enabling real-time detection.

III. MATERIALS AND METHODS

A. Datasets

In this experiment, two different EEG datasets are used. These are:

1) *UCI Epileptic Seizure Dataset*: It is available at Kaggle, contains 500 EEG recordings from 500 subjects, each with 4,097 data points over a duration of 23.6 seconds. It includes five classes: Class 1 (epilepsy), and Classes 2–5 (non epilepsy) covering tumor-affected and healthy brain regions, as well as eyes-open and eyes-closed states.

2) *CHB-MIT Scalp EEG Dataset*: It is available at Physionet, comprises EEG recordings obtained from 23 pediatric subjects at Boston Children's Hospital. It contains 664 EEG files sampled at 256 Hz, with most having 23 channels. For this study, a preprocessed CSV file with 2,097,150 samples across 23 channels was used.

The brief overview of the used dataset for the experiment is shown in Table I.

TABLE I
DESCRIPTION OF THE TWO EEG DATASETS UTILIZED IN THE EXPERIMENT.

Datasets	Data Splitting	No. of EEG Samples	Splitting %
UCI Epileptic Seizure	Training	7360	64%
	Validation	1840	16%
	Testing	2300	20%
CHB-MIT Scalp EEG	Training	1342176	64%
	Validation	335544	16%
	Testing	419230	20%

B. 1D CNN

CNNs are DL-based networks used for feature learning in various tasks. They consist of five layers: input, convolutional, activation, pooling, and fully connected. 1D CNN is used for processing 1D sequence data, processing features through convolutional filters and pooling layers.

C. BiGRU

Gated Recurrent Unit (GRU) is a simpler alternative to LSTM network. Its architecture comprises: input layer, hidden layer, update gate (z_t), reset gate (r_t), candidate activation vector ($\tilde{h}_t$), and output layer (h_t). These components are defined mathematically as follows:

$$r_t = \sigma(W_r x_t + U_{(t-1)} + b_r) \quad (1)$$

$$z_t = \sigma(W_z x_t + U_{(t-1)} + b_z) \quad (2)$$

$$\tilde{h}_t = \tanh(W_h x_t + U_h(r_t \odot h_{(t-1)}) + b_h) \quad (3)$$

$$h_t = (1 - z_t) \odot h_{(t-1)} + z_t \odot \tilde{h}_t \quad (4)$$

Here, W and U are weight matrices, σ is sigmoid activation function, $\odot$ denotes element-wise multiplication, and b is bias vector.

BiGRU is a sequence processing model made up of two GRU - one processes the information in a forward direction while the other does so in reverse using input and forget gates. Its output at time step t is:

$$h_t = \text{concat} \left(\vec{h}_t, \overleftarrow{h}_t \right) \quad (5)$$

where $\overleftarrow{h}_t$ is backward GRU hidden state, and $\vec{h}_t$ is forward GRU hidden state at time step t .

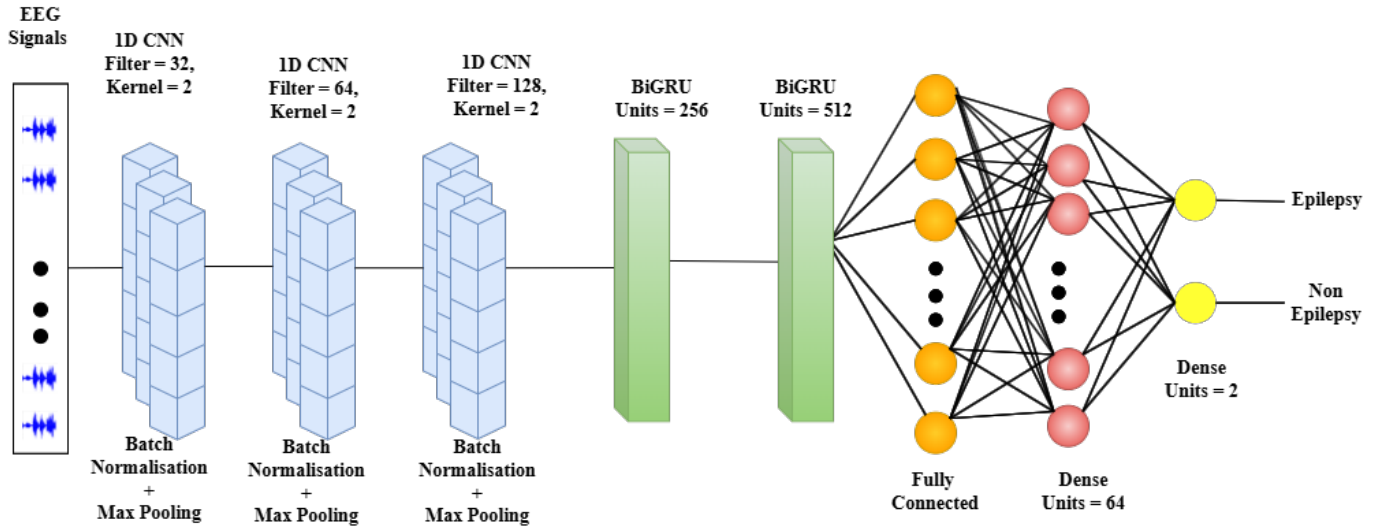


Fig. 1. Detailed architecture of the proposed ESDNet model.

D. Proposed Hybrid Deep Sequential Network Architecture

The proposed ESDNet model is visualized in Figure 1 and described below:

In the input layer, EEG signals are fed into the model. The input shape indicates that the sequence has multiple time steps, with one feature per time step. The 1D convolutional with 32 filters and 2 kernels applied to the input data, which is commonly used for extracting features from sequences, such as time series data. Batch normalisation is applied to normalise the output of the previous 1D CNN layer. It helps in speeding up training and making the model more stable by reducing internal covariate shifts. Subsequently, max-pooling is employed to downsample the feature maps, thereby reducing their dimensionality. Another 1D convolutional layer, this time with 64 filters and 2 kernels followed by batch normalization and max pooling. A third convolutional layer is applied with 128 filters and 2 kernels followed by batch normalization and max pooling. The extracted features are subsequently processed by two stacked BiGRU layers with 256 and 512 units, enabling the capture of long-range temporal dependencies across past and future contexts. Finally, fully connected layers with 64 and 2 units are applied, and a softmax layer produces probability scores for epileptic and non-epileptic classes.

IV. EXPERIMENTAL RESULT AND ANALYSIS

The experiments are performed on two different datasets by taking 64% training, 16% testing and 16% validation. The list of hyperparameters used in this study is shown in Table II. The proposed ESDNet model is compared against existing methods to demonstrate its efficiency in epilepsy detection using two different datasets. Table III reports the accuracy, precision, recall, and F_1 -score, while the confidence interval analysis at 90%, 95%, and 99% confidence levels is reported in Table IV. The proposed ESDNet model achieved 99.74% accuracy with recall, precision, and F_1 -score all equal to 1.0

TABLE II
SUMMARY OF TRAINING HYPER-PARAMETERS OF THE PROPOSED ESDNET MODEL.

Parameters	Values
No of epochs	100 (UCI Epileptic Seizure Dataset), 50 (CHB-MIT Scalp EEG Dataset)
Activation function	ReLU
Batch Size	32 (UCI Epileptic Seizure Dataset), 528 (CHB-MIT Scalp EEG Dataset)
Learning Rate	0.001
Dropout rate	30%
Loss	sparse_categorical_crossentropy
Optimizer	Adam

TABLE III
RESULTS OBTAINED FROM EXPERIMENTS CONDUCTED WITH THE PROPOSED ESDNET MODEL ON TWO DATASETS ARE PRESENTED USING STANDARD CLASSIFICATION EVALUATION MEASURES.

Datasets	Precision	Recall	F_1 -score	Accuracy (%)
UCI Epileptic Seizure	1.00	1.00	1.00	99.74
CHB-MIT Scalp EEG	0.93	0.93	0.93	93.20

on the UCI Epileptic Seizure Dataset, and 93.20% accuracy with corresponding scores of 0.93 on the CHB-MIT Scalp EEG Dataset, demonstrating robust and reliable performance.

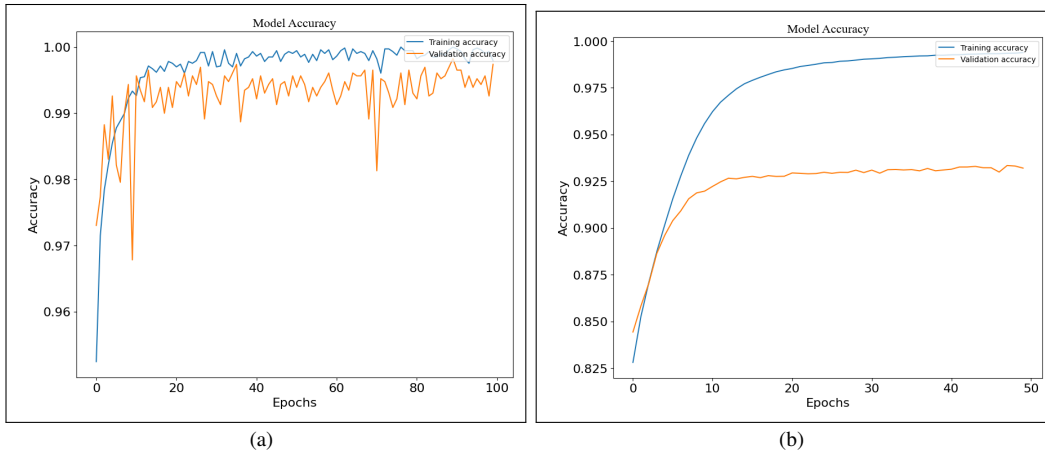


Fig. 2. Training and validation accuracy curve across epochs for (a) UCI Epileptic Seizure Dataset and (b) CHB-MIT Scalp EEG Dataset.

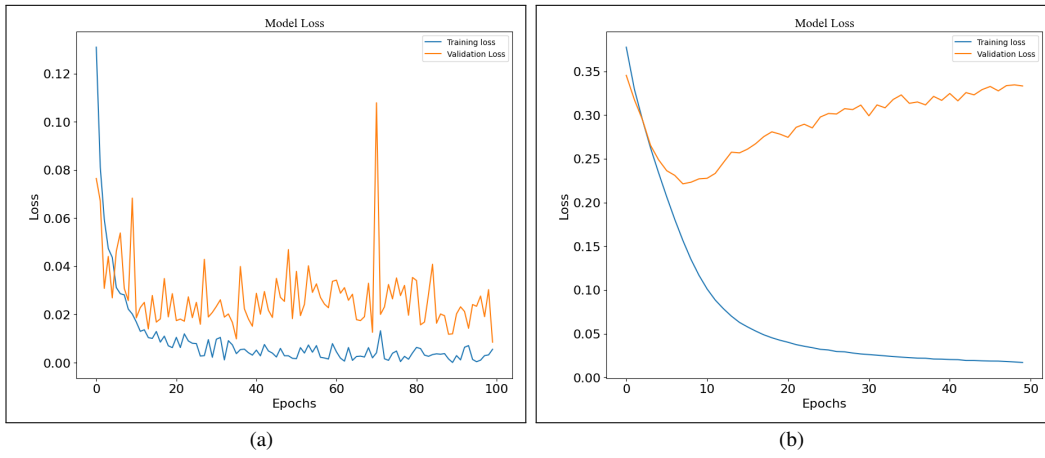


Fig. 3. Training and validation loss curve across epochs for (a) UCI Epileptic Seizure Dataset and (b) CHB-MIT Scalp EEG Dataset.

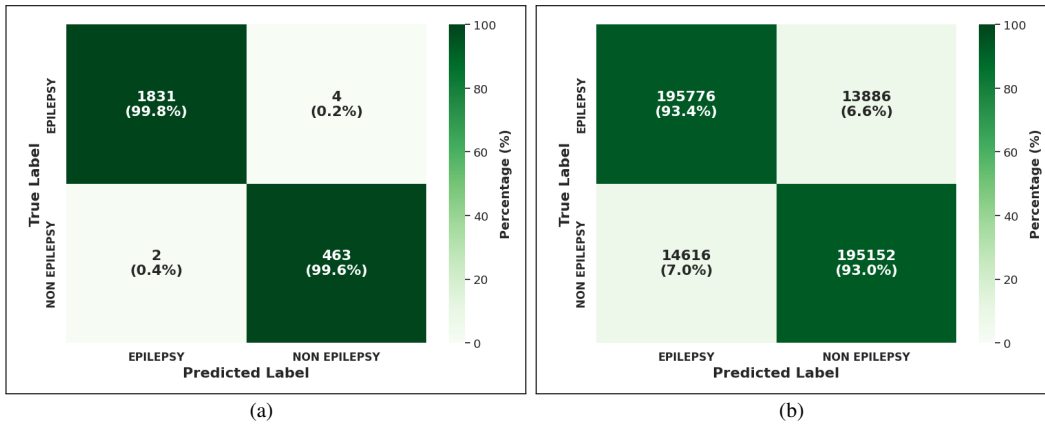


Fig. 4. Confusion matrix of test data for (a) UCI Epileptic Seizure Dataset and (b) CHB-MIT Scalp EEG Dataset.

V. COMPARED METHODS

By incorporating these additional metrics, we have introduced a comprehensive evaluation framework that captures various aspects of the model's effectiveness as shown in

Table V. Our research shown superior percentage accuracy performance. This comparison highlights the novelty and significance of our findings, indicating a substantial improvement over existing approaches.

Table V presents a comparison of classification accuracies

TABLE IV
RESULTS OF CONFIDENCE INTERVAL (CI) ANALYSIS SHOWING THE ERROR RATE AND CORRESPONDING BOUNDS OF THE PROPOSED ESDNET MODEL EVALUATED ON TWO DATASETS AT CONFIDENCE LEVELS (CL) OF 90%, 95%, AND 99%.

Datasets	CL	Confidence Interval ER $\pm$ EB
UCI Epileptic Seizure	90 %	0.003 $\pm$ 0.002
	95 %	0.003 $\pm$ 0.002
	99 %	0.003 $\pm$ 0.003
CHB-MIT Scalp EEG	90 %	0.068 $\pm$ 0.001
	95 %	0.068 $\pm$ 0.001
	99 %	0.068 $\pm$ 0.001

TABLE V
PERFORMANCE COMPARISONS OF THE PROPOSED ESDNET MODEL WITH OTHER EXISTING METHODS IN TERMS OF PERCENTAGE ACCURACY ON TWO DATASETS.

Dataset	Method	Accuracy (%)
UCI Epileptic Seizure	Xavier-PSO-ELM [8]	91.86
	LSTM [9]	97
	1D CNN [10]	99
	Proposed ESDNet	99.74
CHB-MIT Scalp EEG	CNN + Deep Metric Learnings [2]	86.68
	1D CNN + LSTM [11]	90
	CNN [12]	84
	Proposed ESDNet	93.20

between the proposed ESDNet model and several benchmark methods across two widely used EEG datasets. On the UCI Epileptic Seizure dataset, ESDNet attains an accuracy of 99.74%, exceeding the performance of both traditional ML techniques and DL models, including Xavier-PSO-ELM, LSTM, and 1D CNN-based approaches. Likewise, on the CHB-MIT scalp EEG dataset, the proposed ESDNet model achieves an accuracy of 93.20%, outperforming existing CNN- and LSTM-based methods. These outcomes demonstrate the robustness and generalization capability of proposed ESDNet model across heterogeneous EEG datasets.

VI. CONCLUSIONS AND FUTURE WORKS

Epilepsy is a chronic neurological disorder that persists over time and has a significant impact on normal brain function. This study proposes a novel hybrid deep sequential network, termed ESDNet, for epilepsy detection using EEG signals. The proposed ESDNet model employs three 1D CNN layers and two BiGRU layers for effective feature extraction, followed by a dense layer for classification. The proposed ESDNet model is evaluated on two EEG datasets (viz., UCI

Epileptic Seizure and CHB-MIT Scalp) in terms of four evaluations measures (viz., precision, accuracy, recall, F1-score, and CI. Obtained results demonstrate that ESDNet achieves an accuracy of 99.74% on the UCI Epileptic Seizure Dataset and 93.20% on the CHB-MIT Scalp EEG Dataset, outperforming existing benchmark approaches. This empirical comparison highlights the novelty and significance of the proposed ESDNet model, indicating substantial performance improvements over existing advanced methods.

Future work will focus on applying the proposed ESDNet model to additional EEG datasets and exploring federated and transfer learning strategies to enhance generalization.

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